



Analog IC Design ASSESSMENT 2

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PART A: N-MOS

1 Aim

To plot the following characteristics for NMOS:

1. I_D/W v/s g_m/I_D .
2. f_t v/s V_{GS} .

2 Procedure

A) To open Virtuoso Tool

1. Open MobaXterm.
2. In the menu bar, click on Session. Session Setting window will open.
3. Enter your credentials and click on OK.
4. Type the command: `ssh` and click the Enter key.
5. Type the command: `virtuoso &` and click the Enter key. The Virtuoso log window will appear on the screen.

B) To create library

1. In the Virtuoso log window, click on File $\gg$ New $\gg$ Library. A window will appear.
2. Enter the library name, and then click on Attach to an existing technology option and then select UMC_18_CMOS. After selection, click on OK. A library is created.

C) To create cell view

1. In the Virtuoso log window, click on Tools $\gg$ Library Manager.
2. Select the library in which the cell view has to be created.
3. Click on File $\gg$ New $\gg$ Cell View and then click on OK. A cell view is created and the schematic window opens.

D) Circuit Implementation

In the taskbar of the schematic window, click on Create $\gg$ Instance. An instance window appears on the screen. Attach the following instances on the schematic window:

Library Name	Cell Name	Parameters
UMC_18_CMOS	N_18_MM	Total width: 1um Finger width: 1um Total length: L m Finger number: 1
analogLib	vdc	DC voltage: V_{GS}
analogLib	vdc	DC voltage: V_{DS}
analogLib	gnd	—

Circuit Diagram

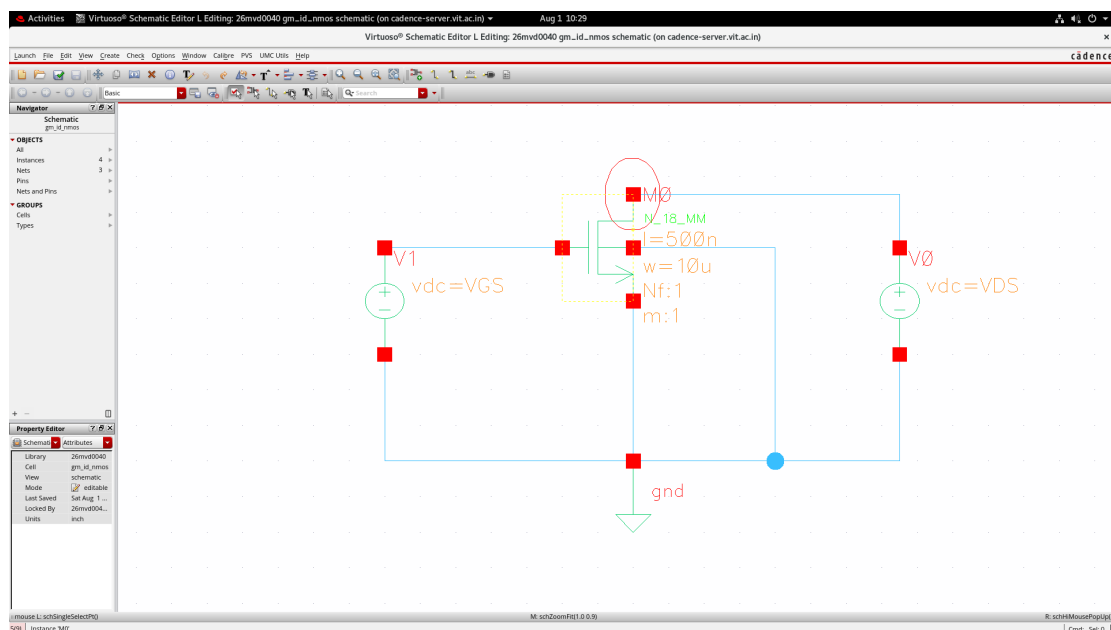


Figure 1: Circuit Diagram for NMOS.

To connect all the placed components in the schematic window, click on Create in the taskbar of the schematic window and then select Wire(narrow). Connect all components as shown in Figure 1. After connecting all components, click the Check & Save button and check for any errors in the circuit which will appear in the log window.

3 Simulation Procedure

A) To open Analysis window

Click on the Launch button present in the taskbar of the schematic window and then select ADE_L. The analysis window opens.

B) To plot I_D/W vs g_m/I_D for varying length

1. In the analysis window, click on the Setup button. Then click on Model Libraries. A dialog window appears, check whether the library mm180_reg18-v124.lin-scs has the tt option. Click on OK.
2. In the analysis window, select the Analyses option in the taskbar. Then select the Choose option. A window appears.
3. Select dc, Save operating point option and ensure the Enabled option is toggled on. Click OK.
4. In the analysis window, select the Variable option in the taskbar and select the Copy from cell view option. The variables in the circuit schematic appear in the Variables section of the analysis window. Enter:

$$L = 180n, \quad V_{DS} = 100m, \quad V_{GS} = 0.$$

5. Double click on the created analysis in the analysis window. Enable Design Variable, click on Select Design Variable and select V_{GS} . Enter:

$$\text{Start} = 0, \quad \text{Stop} = 1.8.$$

6. In the analysis window, click on Output $\gg$ To be plotted and then select Select on Design. Select the drain terminal in the schematic window. Make sure only the drain terminal is selected in the Output section.
7. Select the Simulation option in the taskbar and click on Netlist & Run. The simulation window opens with the plot of I_D v/s V_{GS} .
8. Go to the MobaXterm terminal and type: gedit saveop.scs &. A file appears. Type:

```
save m0:all
save
```

and close the file.

9. In the analysis window, select Setup and then select Simulation Files. Click on Stimulus Files, select the saveop.scs file and click on OK.
10. In the analysis window, select Tools and then select Results Browser. Select the following and send them to the Calculator.

Parameter	Source to fetch the parameter
g_m	dc $\gg$ Signals $\gg$ gm
I_D	dc $\gg$ Signals $\gg$ gm
w	instance $\gg$ M0 $\gg$ Signals $\gg$ w

11. In the Calculator, create the following expressions:

```
(getData("M0:gm" ?result "dc") / getData("M0:id" ?result "dc"))
```

```
(getData("M0:id" ?result "dc") / pv("M0" "w" ?result "instance"))
```

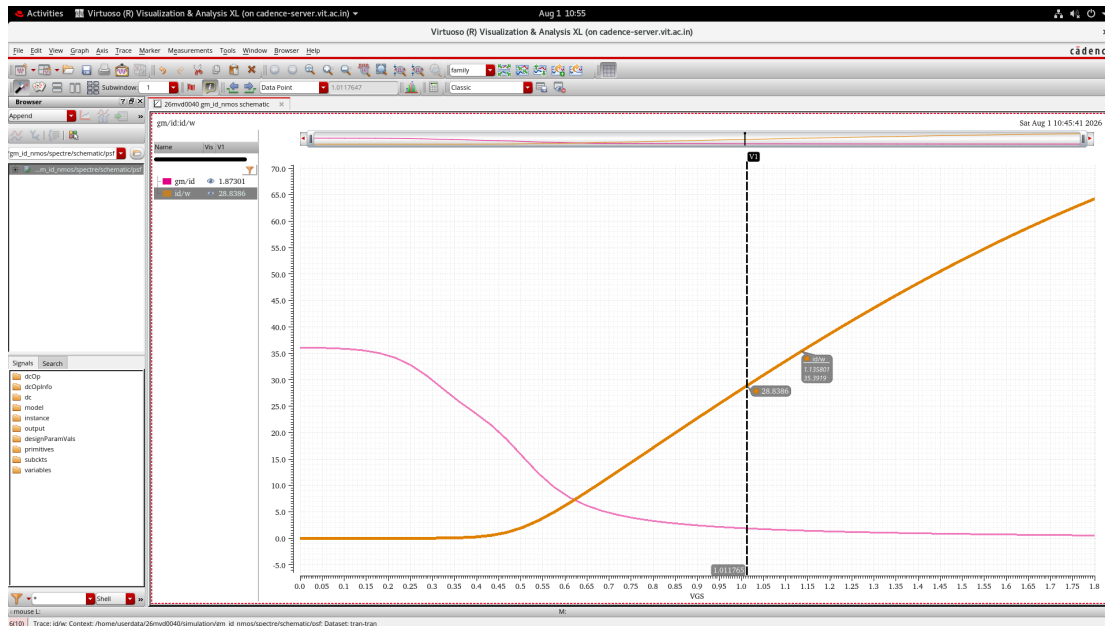


Figure 2: g_m/i_d and i_d/w plgraph otted in the same plot for NMOS.

12. After creating each expression in the Calculator, go to Outputs in the analysis window, then select Setup. Select New Expression and then select Get Expression. Give the expression a name and click on OK.
13. In the analysis window, select Tools $\gg$ Calculator. In the function window of the Calculator search “wavevsWave” and select the wavevsWave function. Enter:

x trace:

```
(getData("M0:gm" ?result "dc") / getData("M0:id" ?result "dc"))
```

y trace:

```
(getData("M0:id" ?result "dc") / pv("M0" "w" ?result "instance"))
```

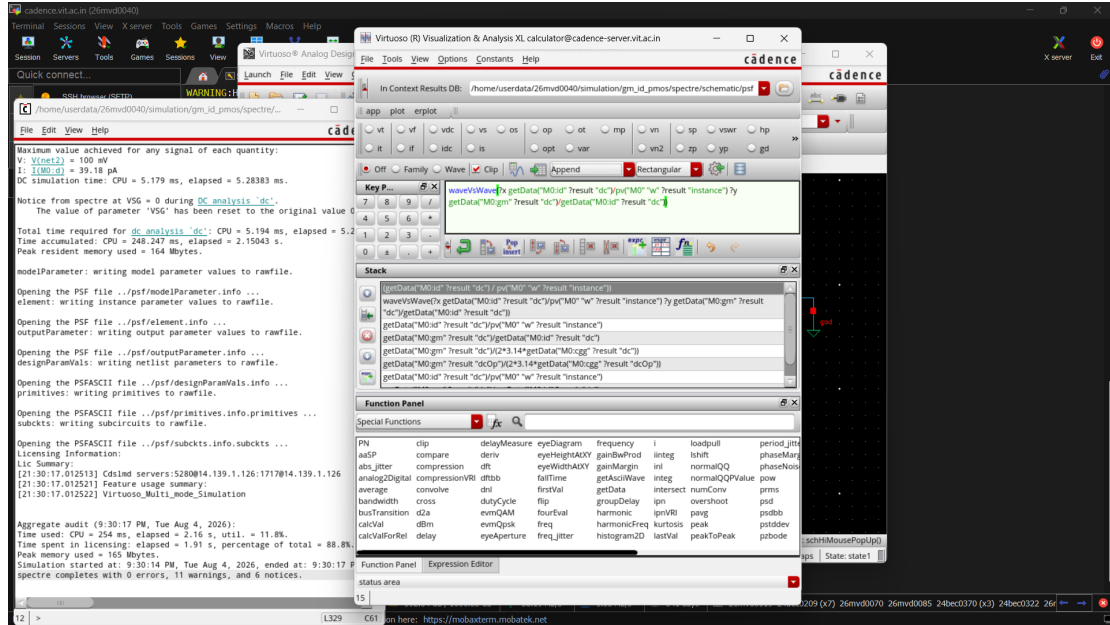


Figure 3: Calculator.

14. After selecting the parameters, click on OK. An expression appears in the Calculator. Bring this expression to the Outputs section of the analysis window.
15. In the taskbar of the analysis window, select Tools and then Parametric option. The parametric analysis window opens. Give:

Select Variable	Start	Stop
L	180n	2u

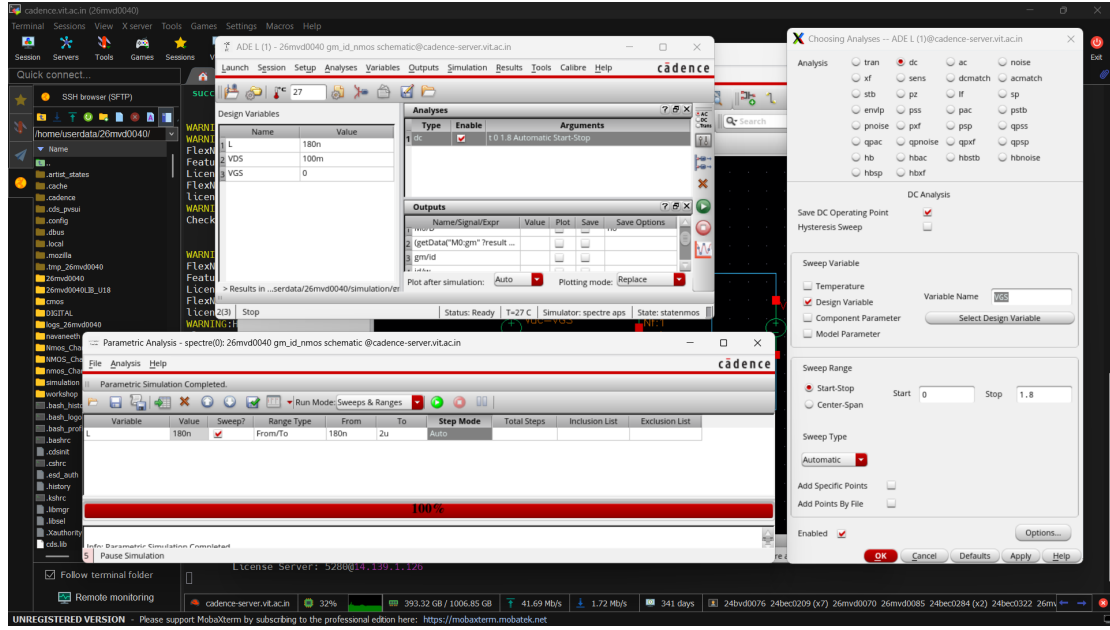


Figure 4: Settings for plotting I_D/W v/s g_m/I_D for NMOS.

16. The simulation window opens with the plot of I_D/W v/s g_m/I_D for varying length as shown in Figure 5.

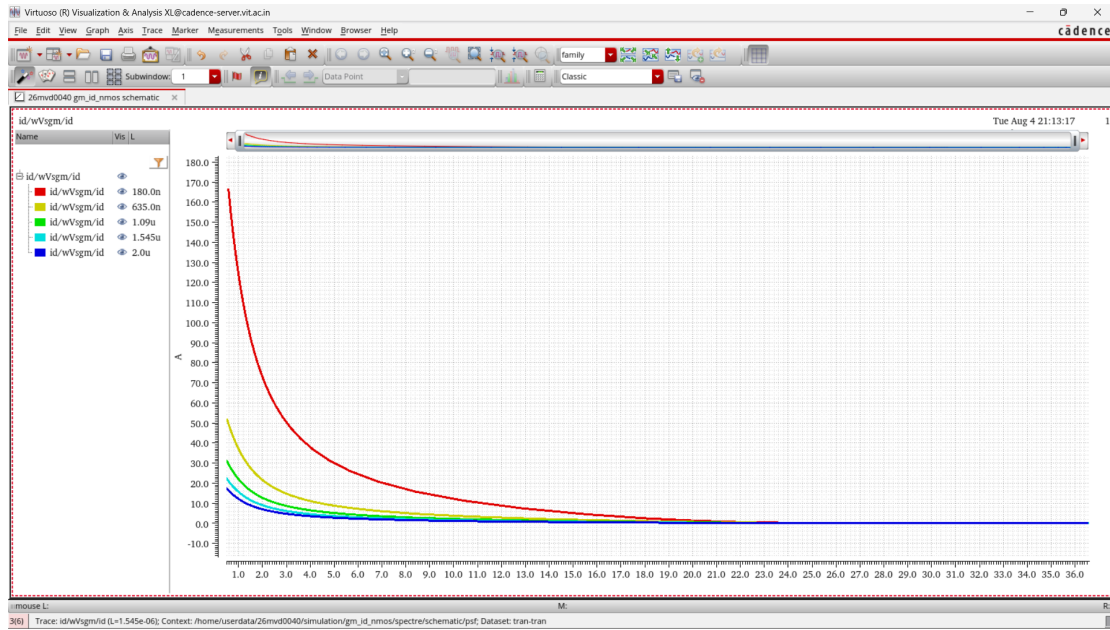


Figure 5: I_D/W v/s g_m/I_D for varying length plot for NMOS.

C) To plot f_t vs V_{GS}

1. In the analysis window, select Tools and then Results Browser. Select the following and send them to the Calculator:

Parameter	Source to fetch the parameter
g_m	dc >> Signals >> gm
I_D	dc >> Signals >> gm
c_{gg}	dc >> Signals >> cgg

2. In the Calculator, create:

```
(getData("M0:gm" ?result "dc")/(2*3.14*getData("M0:cgg" ?result "dc")))
```

3. After creating the expression in the Calculator, go to Outputs in the analysis window, then select Setup. Select New Expression and then Get Expression. Give the expression a name and click on OK.
4. Finally select the Simulation option in the taskbar and click on Netlist & Run. The simulation window opens with the plot of f_t v/s V_{GS} .

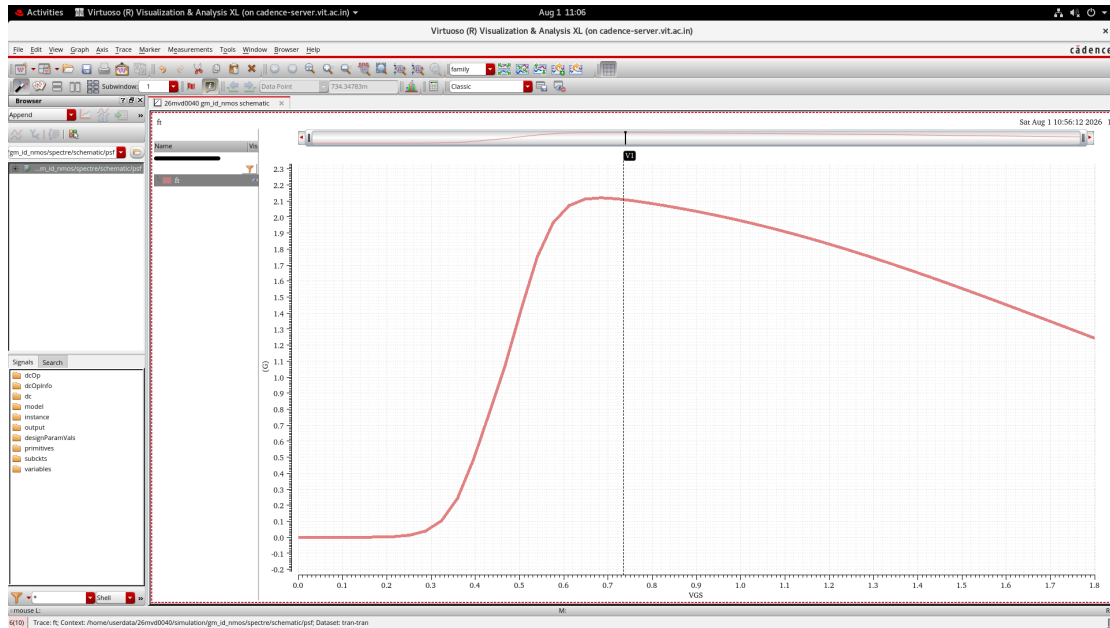


Figure 6: f_t v/s V_{GS} plot for NMOS.

4 Observations

- A) I_D/W v/s g_m/I_D plot: As g_m/I_D increases, I_D/W decreases exponentially. Shorter channel lengths yield significantly higher I_D/W value.
- B) f_t v/s V_{GS} plot: f_t rises sharply as V_{GS} increases, reaches a maximum peak and then gradually decreases.

Result

All the curves are plotted and observations from the plots are inferred successfully.

PART B: P-MOS

5 Aim

To plot the following characteristics for PMOS:

1. I_D/W v/s g_m/I_D .
2. f_t v/s V_{GS} .

6 Procedure

A) To open Virtuoso Tool

1. Open MobaXterm.
2. In the menu bar, click on Session. Session Setting window will open.
3. Enter your credentials and click on OK.
4. Type the command: `ssh` and click the Enter key.
5. Type the command: `virtuoso &` and click the Enter key. The Virtuoso log window will appear on the screen.

B) To create library

1. In the Virtuoso log window, click on File $\gg$ New $\gg$ Library. A window will appear.
2. Enter the library name, and then click on Attach to an existing technology option and then select UMC_18_CMOS. After selection, click on OK. A library is created.

C) To create cell view

1. In the Virtuoso log window, click on Tools $\gg$ Library Manager.
2. Select the library in which the cell view has to be created.
3. Click on File $\gg$ New $\gg$ Cell View and then click on OK. A cell view is created and the schematic window opens.

D) Circuit Implementation

In the taskbar of the schematic window, click on Create $\gg$ Instance. An instance window appears on the screen. Attach the following instances on the schematic window:

Library Name	Cell Name	Parameters
UMC_18_CMOS	P_18_MM	Total width: 1um Finger width: 1um Total length: L m Finger number: 1
analogLib	vdc	DC voltage: V_{GS}
analogLib	vdc	DC voltage: V_{DS}
analogLib	vdc	DC voltage: V_{BS}
analogLib	gnd	—

Circuit Diagram

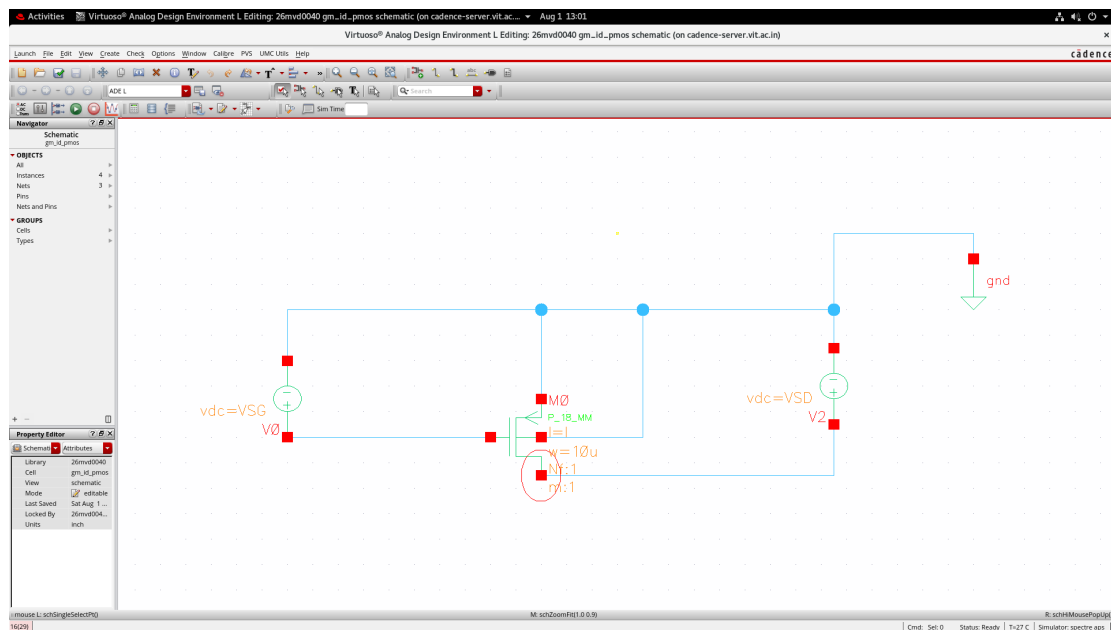


Figure 7: Circuit Diagram for PMOS.

To connect all the placed components in the schematic window, click on Create in the taskbar of the schematic window and then select Wire(narrow). Connect all components as shown in Figure 4. After connecting all components, click the Check & Save button and check for any errors in the circuit which will appear in the log window.

7 Simulation Procedure

A) To open Analysis window

Click on the Launch button present in the taskbar of the schematic window and then select ADE_L. The analysis window opens.

B) To plot I_D/W vs g_m/I_D for varying length

1. In the analysis window, click on the Setup button. Then click on Model Libraries. A dialog window appears, check whether the library mm180_reg18-v124.lin-scs has the tt option. Click on OK.
2. In the analysis window, select the Analyses option in the taskbar. Then select the Choose option. A window appears.
3. Select dc, Save operating point option and ensure the Enabled option is toggled on. Click OK.
4. In the analysis window, select the Variable option in the taskbar and select the Copy from cell view option. Enter:

$$L = 180n, \quad V_{DS} = -100m, \quad V_{GS} = 0.$$

5. Double click on the created analysis in the analysis window. Enable Design Variable, click on Select Design Variable and select V_{GS} . Enter:

$$\text{Start} = -1.8, \quad \text{Stop} = 0.$$

6. In the analysis window, click on Output $\gg$ To be plotted and then select Select on Design. Select the drain terminal in the schematic window. Make sure only the drain terminal is selected in the Output section.
7. Select the Simulation option in the taskbar and click on Netlist & Run. The simulation window opens with the plot of I_D v/s V_{GS} .
8. Go to the MobaXterm terminal and type: gedit saveop.scs &. A file appears. Type:

```
save m0:all
save
```

and close the file.

9. In the analysis window, select Setup and then select Simulation Files. Click on Stimulus Files, select the saveop.scs file and click on OK.
10. In the analysis window, select Tools and then select Results Browser. Select the following and send them to the Calculator.

Parameter	Source to fetch the parameter
g_m	dc >> Signals >> gm
I_D	dc >> Signals >> gm
w	instance >> M0 >> Signals >> w

11. In the Calculator, create:

```
(getData("M0:gm" ?result "dc") / getData("M0:id" ?result "dc"))
```

```
(getData("M0:id" ?result "dc") / pv("M0" "w" ?result "instance"))
```

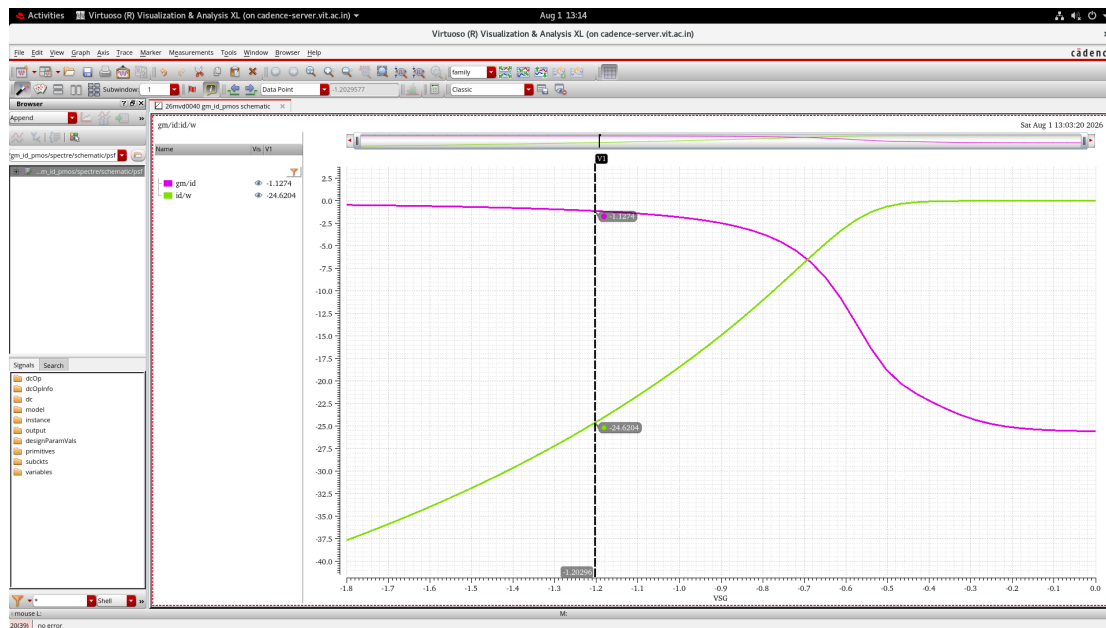


Figure 8: g_m/id and id/w plotted in the same plot for PMOS.

12. After creating each expression in the Calculator, go to Outputs in the analysis window, then select Setup. Select New Expression and then Get Expression. Give the expression a name and click on OK.
13. In the analysis window, select Tools >> Calculator. In the function window of the Calculator search “wavevsWave” and select the wavevsWave function. Enter:

x trace:

```
(getData("M0:gm" ?result "dc")/getData("M0:id" ?result "dc"))
```

y trace:

```
(getData("M0:id" ?result "dc")/pv("M0" "w" ?result "instance"))
```

14. After selecting the parameters, click on OK. An expression appears in the Calculator. Bring this expression to the Outputs section of the analysis window.
15. In the taskbar of the analysis window, select Tools and then Parametric option. The parametric analysis window opens. Give:

Select Variable	Start	Stop
L	180n	2u

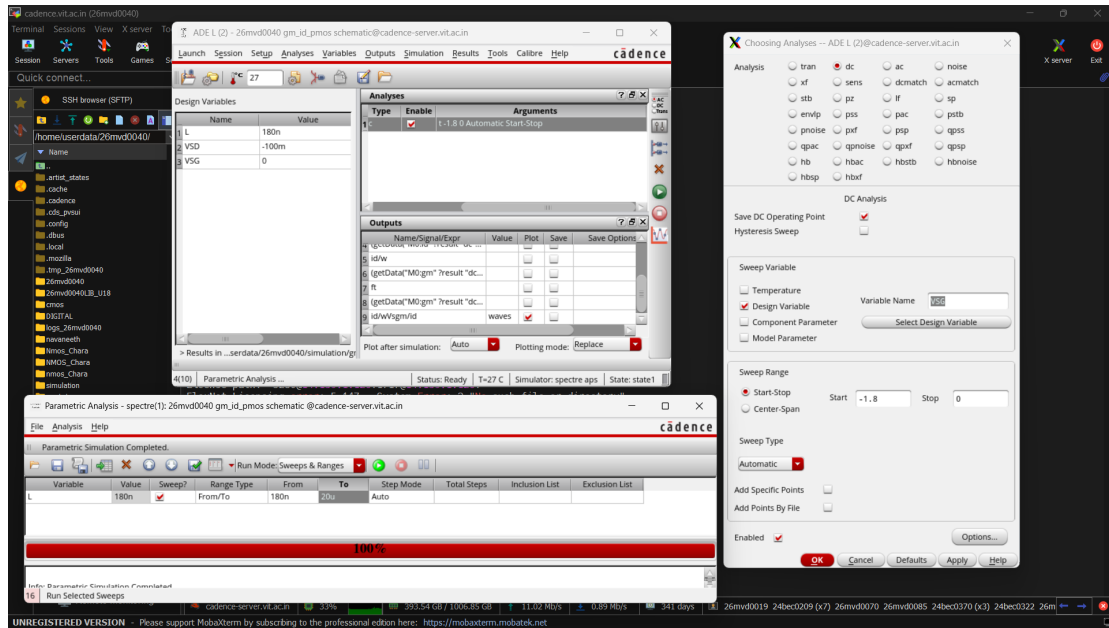


Figure 9: Settings for plotting I_D/W v/s g_m/I_D for PMOS.

16. The simulation window opens with the plot of I_D/W v/s g_m/I_D for varying length as shown in Figure 10.

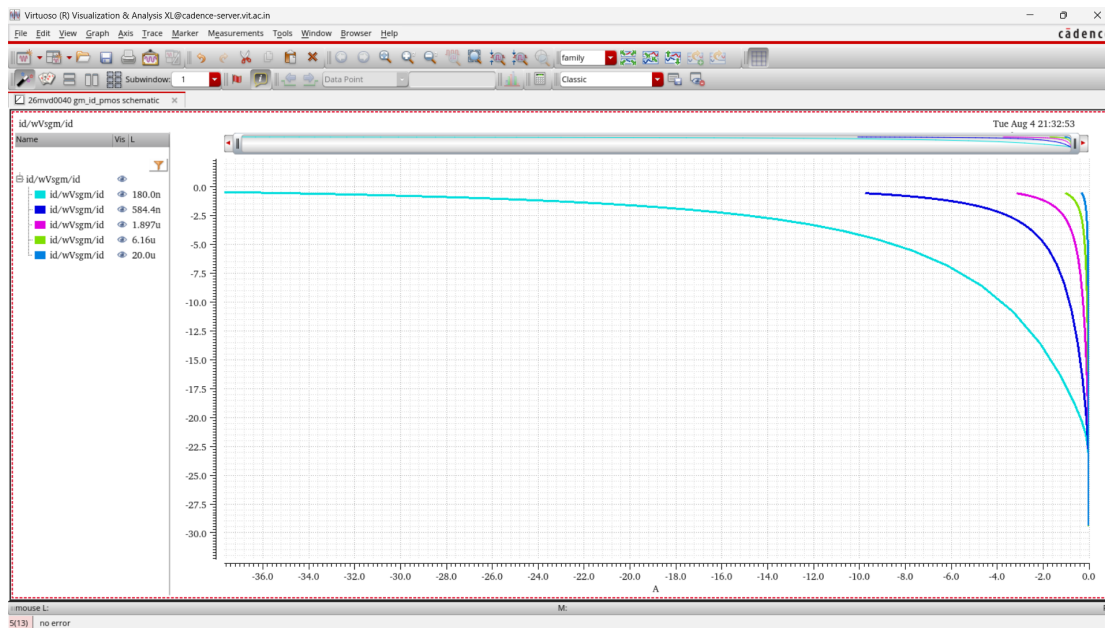


Figure 10: I_D/W v/s g_m/I_D for varying length plot for PMOS.

C) To plot f_t vs V_{GS}

1. In the Calculator, create the following expression:

```
waveVsWave(?x abs(pv("/top-level" "VGS" ?result "designParamVals"))
?y (getData("M0:gm" ?result "dc") /
(2 * 3.14 * getData("M0:cgg" ?result "dc"))))
```

2. After creating the expression in the Calculator, go to Outputs in the analysis window, then select Setup. Select New Expression and then Get Expression. Give the expression a name and click on OK. The expression appears on the Output section of the analysis window.
3. Finally select the Simulation option in the taskbar and click on Netlist & Run. The simulation window opens with the plot of f_t v/s V_{GS} .

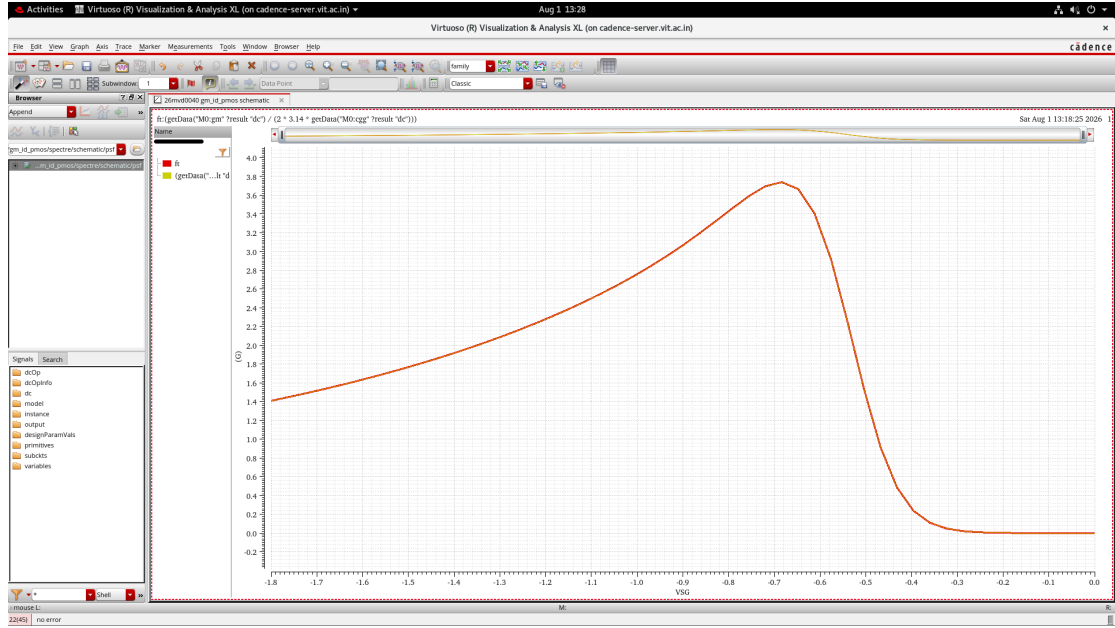


Figure 11: f_t v/s V_{GS} plot for PMOS.

8 Observations

- A) I_D/W v/s g_m/I_D plot: As g_m/I_D increases, I_D/W decreases exponentially. Shorter channel lengths yield significantly higher I_D/W value.
- B) f_t v/s V_{GS} plot: f_t rises sharply as V_{GS} increases, reaches a maximum peak and then gradually decreases.

Result

All the curves are plotted and observations from the plots are inferred successfully.